XIAMEN UNIVERSITY MALAYSIA

TEST QUESTIONS (20%)

Course Code:	COM106		
Course Name:	Electrical Engineering		
Question Paper Setter:	Assoc. Prof. Ir. Ts. Dr. Wong Shen Yuong		
Academic Session:	2022/04	Question Paper:	A ☑
Total No. of Pages:	2	Time Allocated:	1 hour
Additional Materials:	-		
Apparatus Allowed:	Calculator		

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 3 Questions.
   Answer **ALL** questions.
2. Read the above information carefully to ensure you have the correct and complete question paper.
3. Please follow the requirement of each section and write down all the corresponding answers on the lined paper.
4. Only PDF hand written answer sheet will be accepted.
5. Submit your answer sheet on Moodle link " Submission", which there will be a cut-off time.

DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.

(Student ID: _____ Full Name: _____)

Question 1 [3 marks]

(CLO 1, PLO 1, C2)

(a) Explain Kirchhoff's voltage law. [1 mark]

(b) Name TWO (2) subdivisions of Electrical Engineering. [2 mark]

Question 2 [9 Marks]:

(CLO 1, PLO 1, C3)

(a) Find *V* in the circuit of Figure 1 using KVL. [3 marks]

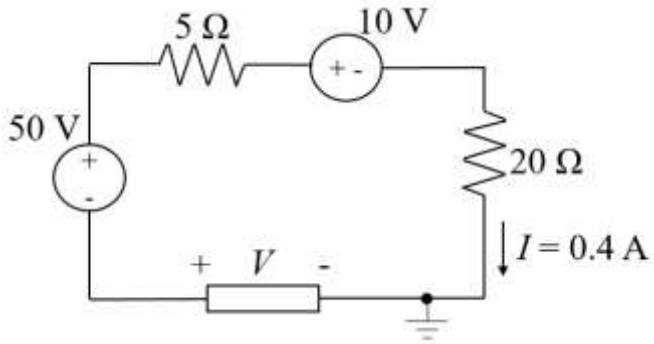


Figure 1

(b) Calculate the total resistance R_{eq} and calculate voltage *V* and current *I* that flow across 8Ω resistor in the circuit of Figure 2. [6 marks]

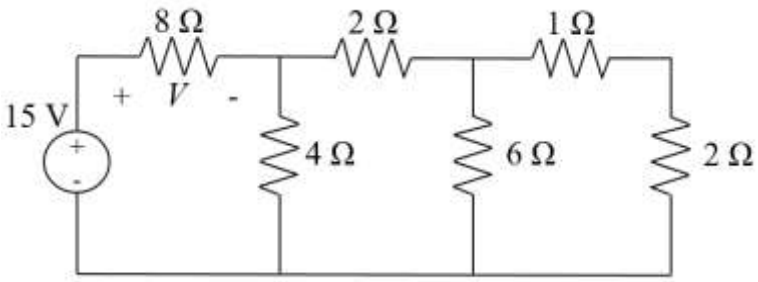


Figure 2

Question 3 [8 Marks]:

(CLO 1, PLO 1, C3)

Referring to Figure 3,

(a) Write the network equations for node voltages V_1 and V_2. [2 marks]

(b) Put the network equations into standard form and matrix. [2 marks]

(c) Solve for V_1 and V_2. [4 marks]

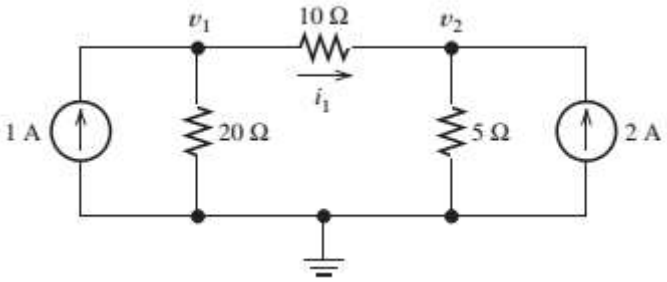


Figure 3

- **End of Question -**